

ROOTS OF EQUATIONS

1. ♦ Write the steps of Bisection Method. (2025-26)
- ♦ Explain in brief about bisection method. (2024-25)
- ♦ Explain the bisection method. Find the root of the equation $x^3 - x - 11 = 0$. (2023-24)
- ♦ Explain Bisection method for finding root of the equation. (2018)
- ♦ Prove that Bisection method always converges.

Bisection Method

Let $[p_n, q_n]$ be the interval at n^{th} step of bisection, having a root of the equation $f(x) = 0$.

Let x_n be the n^{th} approximation for the root. Then initially $p_1 = a$ and $q_1 = b$.

$$\Rightarrow x_1 = \text{first approximation} = \left(\frac{p_1 + q_1}{2} \right)$$

$$\Rightarrow p_1 < x_1 < q_1$$

Now either the root lies in $[a, x_1]$ or in $[x_1, b]$.

$$\therefore \text{either } [p_2, q_2] = [p_1, x_1] \text{ or } [p_2, q_2] = [x_1, q_1]$$

$$\Rightarrow \text{either } p_2 = p_1, q_2 = x_1 \text{ or } p_2 = x_1, q_2 = q_1$$

$$\Rightarrow p_1 \leq p_2, q_2 \leq q_1$$

$$\text{Also, } x_2 = \frac{p_2 + q_2}{2} \text{ so that } p_2 < x_2 < q_2$$

Continuing this way, we obtain that at n^{th} step,

$$x_n = \frac{p_n + q_n}{2}, p_n < x_n < q_n$$

And $p_1 \leq p_2 \leq \dots \leq p_n$ and $q_1 \geq q_2 \geq \dots \geq q_n$

$\therefore < p_1, p_2, \dots, p_n, \dots >$ is bounded non-decreasing sequence bounded by b and $< q_1, q_2, \dots, q_n, \dots >$ is a bounded non-increasing sequence of numbers bounded by a .

Hence both these sequences converge.

Let, $\lim_{n \rightarrow \infty} p_n = p$ and $\lim_{n \rightarrow \infty} q_n = q$.

Now since length of the interval is decreasing at every step, we get that

$$\lim_{n \rightarrow \infty} (q_n - p_n) = 0 \Rightarrow q = p$$

$$\text{Also, } p_n < x_n < q_n$$

$$\Rightarrow \lim p_n \leq \lim x_n \leq \lim q_n$$

$$\Rightarrow p \leq \lim x_n \leq q$$

$$\Rightarrow \lim x_n = p = q$$

Further since a root lies in $[p_n, q_n]$, we shall have

$$f(p_n)f(q_n) < 0$$

$$\Rightarrow 0 \geq \lim_{n \rightarrow \infty} [f(p_n)f(q_n)]$$

$$\Rightarrow 0 \geq f(p)f(q)$$

$$\Rightarrow 0 \geq [f(p)]^2$$

Example : Root of the $x^3 - x - 11 = 0$

Now first finding the interval at value lie

So

$$f(x) = x^3 - x - 11$$

$$f(2) = (2)^3 - 2 - 11 = -5$$

$$f(3) = (3)^3 - 3 - 11 = 13$$

So the root lies between the interval (2 - 3).

Starting with $x_1 = 2$ and $x_2 = 3$

$$x_3 = \frac{2+3}{2} = 2.5$$

$$f(x_3) = f(2.5) = (2.5)^3 - 2.5 - 11 = 2.125$$

$$\text{Now } f(x_1) \times f(x_3) = -5 \times 2.125 = -10.625 < 0$$

Therefore replace x_2 by x_3 . The new search interval (2, 2.5)

$$x_3 = \frac{2+2.5}{2} = 2.25$$

$$f(x) = f(2.25) = (2.25)^3 - 2.25 - 11 = -1.859375$$

$$\text{Now } f(x_1) \times f(x_3) = -5 \times -1.859375 = 9.296875 > 0$$

The first approximation to root = 2.25. The new search interval is (2.25, 2.5).

$$x_3 = \frac{2.25 + 2.5}{2} = 2.375$$

$$f(2.375) = (2.375)^3 - 2.375 - 11 = 0.021484375$$

So the second approximation root = 2.375 and it is correct to three decimal places.

2. ♦ *Explain in brief about Falsi method. Solve $x^3 - 9x + 1 = 0$ for the root lying between 2 and 4 by Falsi method.* (2025-26)

♦ *Solve $x^3 - 9x + 1 = 0$ for the root lying between 2 and 4 by Regula falsi method.* (2017, 2021, 2024-25)

Falsi Method (Regular Falsi Method)

The Regula Falsi method (or False Position Method) is an iterative numerical technique used to find real roots of a function $f(x) = 0$.

It works by drawing a straight line (chord) between two points $(a, f(a))$ and $(b, f(b))$ where the function changes sign, and using the intersection of this line with the x-axis as an approximation for the root.

$$f(x) = x^3 - 9x + 1, a = 2 \text{ and } b = 4$$

$$f(a) = f(2) = 2^3 - 9 \times 2 + 1 = -9 < 0 \text{ and}$$

$$f(b) = f(4) = 4^3 - 9 \times 4 + 1 = 29 > 0$$

The first approximation to the root is :

$$x_1 = \frac{af(b) - bf(a)}{f(b) - f(a)} = \frac{2 \times 29 - 4(-9)}{29 - (-9)} = 2.474$$

Now

$$f(x_1) = f(2.474) = (2.474)^3 - 9 \times 2.474 + 1 = -6.123 < 0$$

Therefore, the root lies between $x_1 = 2.474$ and $b = 4$

$$\begin{aligned} \text{Now, } x_2 &= \frac{2.474(29) - 4(-6.123)}{29 - (-6.123)} \\ &= 2.740 \end{aligned}$$

$$f(x_2) = f(2.74) = -3.089 < 0$$

Root lies between $x_2 = 2.74$ & $b = 4$

Third approximation

$$x_3 = \frac{2.74(29) - 4(-3.089)}{29 - (-3.089)} = 2.861$$

$$f(x_3) = f(2.861) = -1.331 < 0$$

Therefore root lies between $x_3 = 2.861$ and $b = 4$

$$x_4 = \frac{2.861(29) - 4(-1.331)}{29 - (-1.331)} = 2.911$$

The required root is 2.911.

(Ans.)

3. *Derive the Newton - Raphson method from the Taylor's series. Use Newton-Raphson method to find a root of $f(x) = x^3 - x - 1 = 0$. taking $x_0 = 1$, correct up to three decimal places.*

(2025-26)

Newton - Raphson Method derived from the Taylor's Series

Let x_0 be an approximation root of the equation $f(x) = 0$. Therefore, expanding $f(x_0 + h)$ by Taylor series we get;

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \frac{h^3}{3!} f'''(x_0) + \dots + \frac{h^n}{n!} f^{(n)}(x_0) + O(h^{n+1}) = 0$$

Since h is small, neglecting h^2 and above terms

$$f(x_0) + hf'(x_0) = 0$$

$$h = -\frac{f(x_0)}{f'(x_0)}$$

Therefore, a closure approximation to the root is given by

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Similarly, starting with x_1 a still better approximation x_2 is given by

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

In general

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \text{ which is known as newton raphson}$$

formula

Hence Derived.

Numerical Solution

Here $x^3 - x - 1 = 0$

Let $f(x) = x^3 - x - 1$

$$\frac{d}{dx}(x^3 - x - 1) = 3x^2 - 1$$

$$\therefore f'(x) = 3x^2 - 1$$

$$x_0 = 1$$

1st Iteration :

$$f(x_0) = f(1) = 1^3 - 1 - 1 = -1$$

$$f'(x_0) = f'(1) = 3 \cdot 1^2 - 1 = 2$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = 1 - \frac{-1}{2}$$

$$x_1 = 1.5$$

2nd Iteration :

$$f(x_1) = f(1.5) = 1.5^3 - 1.5 - 1 = 0.875$$

$$f'(x_1) = f'(1.5) = 3 \cdot 1.5^2 - 1.5 - 1 = 5.75$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 1.5 - \frac{0.875}{5.75}$$

$$x_2 = 1.3478$$

3rd Iteration :

$$f(x_2) = f(1.3478) = (1.3478)^3 - 1.3478 - 1 = 0.1007$$

$$f'(x_2) = f'(1.3478) = 3(1.3478)^2 - 1 = 4.4499$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$x_2 = 1.3478 - \frac{0.1007}{4.4499}$$

$$x_3 = 1.3252$$

4th Iteration :

$$f(x_3) = f(1.3252) = 1.3252^3 - 1.3252 - 1 = 0.0021$$

$$f'(x_3) = f'(1.3252) = 3(1.3252)^2 - 1 = 4.2685$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$x_4 = 1.3252 - \frac{0.0021}{4.2685}$$

$$x_4 = 1.3247$$

Approximate root of the equation $x^3 - x - 1 = 0$ using Newton Raphson method is 1.3247 (After 4 iterations)

n	x_0	$f(x_0)$	$f'(x_0)$	x_1	Update
1	1	-1	2	1.5	$x_0 = x_1$
2	1.5	0.875	5.75	1.3478	$x_0 = x_1$
3	1.3478	0.1007	4.4499	1.3252	$x_0 = x_1$
4	1.3252	0.0021	4.2685	1.3247	$x_0 = x_1$

Hence; Upto 3 decimal place root = 1.3252

4. ♦ **Derive equation for Newton-Raphson method and discuss its convergence.** (2024-25)
- ♦ **Write the algorithm of Newton-Raphson Method. Using Newton-Raphson method, find the real root of the equation $3x = \cos x + 1$ correct to four decimal places.** (2023-24)
- ♦ **Discuss rate of convergence of Newton - Raphson method.** (2021)
- ♦ **Obtain the rate of convergence for Newton-Raphson method.**

Newton-Raphson Method

This method starts with an initial approximation to the root of the equation. A better and closer approximation to the root can be found by using an iterative process.

Newton - Raphson Formula :

$$X_{i+1} = X_i - \left[\frac{f(X_i)}{f'(X_i)} \right], i = 0, 1, 2, \dots$$

This method can be derived directly by the Taylor expansion $f(x)$ in the neighbourhood of the root ϵ of $f(x) = 0$. The starting approximation x_0 to ϵ is to be properly chosen so that the first order Taylor series approximation of $f(x_0+h)$ in the neighbourhood of x_0 leads to x_1 , an improved approximation to ϵ . i.e.

$$f(x_0+h) = f(x_0) + hf'(x_0) + \frac{h^2}{2}f''(x_0) + \dots = 0$$

$\therefore h \ll 1$ neglecting h^2 and its higher powers, we get

$$f(x_0) + hf'(x_0) = 0$$

i.e.

$$h = -\frac{f(x_0)}{f'(x_0)}, \quad \therefore h = x_1 - x_0$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Now the successive approximations $x_2, x_3, \dots, x_n, \dots$ etc may be calculated by the iterative formula :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Proof of Quadratic Convergence for Newton's Iterative Method

According to Taylor's theorem, any function $f(x)$ which has a continuous second derivative can be represented by an expansion about a point that is close to a root of $f(x)$. Suppose this root is α . Then the expansion of $f(\alpha)$ about x_n is :

$$f(\alpha) = f(x_n) + f'(x_n)(\alpha - x_n) + R_1 \quad \dots(1)$$

Where the Lagrange form of the Taylor series expansion remainder is

$$R_1 = \frac{1}{2!} f''(\xi_n)(\alpha - x_n)^2 ; \text{ Where } \xi_n \text{ is in between } x_n$$

and α

Since α is the root, (1) becomes,

$$0 = f(\alpha) = f(x_n) + f'(x_n)(\alpha - x_n) + \frac{1}{2}f''(\xi_n)(\alpha - x_n)^2 \dots (2)$$

Dividing equation (2) by $f'(x_n)$ and rearranging gives

$$\frac{f(x_n)}{f'(x_n)} + (\alpha - x_n) = \frac{-f''(\xi_n)}{2f'(x_n)}(\alpha - x_n)^2 \dots (3)$$

Remembering that x_{n+1} is defined by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \dots (4)$$

One finds that

$$\underbrace{\alpha - x_{n+1}}_{\epsilon_{n+1}} = \frac{-f''(\xi_n)}{2f'(x_n)} \underbrace{(\alpha - x_n)^2}_{\epsilon_n^2}$$

That is,

$$\epsilon_{n+1} = \frac{-f''(\xi_n)}{2f'(x_n)} \epsilon_n^2 \dots (5)$$

Taking absolute value of both sides gives,

$$[\epsilon_{n+1}] = \frac{|f''(\xi_n)|}{2|f'(x_n)|} \epsilon_n^2 \dots (6)$$

Equation (6) shows that the rate of convergence is quadratic.

Solution of $3x = \cos x + 1$

Here we have,

$$f(x) = 3x - \cos x - 1$$

$$\text{Now, } f'(x) = 3 + \sin x$$

By graphical method, we can obtain the approximation value of the root is 0.61.

$$\text{Now, } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{3x_n - \cos x_n - 1}{3 + \sin x_n}$$

Now, taking $x_0 = 0.61$ we have :

$$\begin{aligned} x_1 &= 0.61 - \frac{0(0.61) - \cos(0.61) - 1}{3 + \sin(0.61)} \\ &= 0.61 - 0.00290 = 0.6071 \end{aligned}$$

$$\text{And } x_2 = 0.6071 - \frac{3(0.6071) - \cos(0.61) - 1}{3 + \sin(0.6071)}$$

$$= 0.61 - 0.00290 = 0.6071 = 0.60710$$

The value of the root is 0.6071.

/*Program to implement Newton-Raphson method

to find root of equation $3x - \cos x - 1 = 0$ */

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#include<math.h>
```

```
double m(double x)
```

```
{
```

```
return 3 * x - cos((double) x) - 1=0;
```

```
}
```

```
double dm(double x)
```

```
{
```

```
return 3 + sin((double) x);
```

```
void main( )
```

```
{
```

```
float p, r, temp, delta, reller;
```

```
int i, n;
```

```
clrscr( );
```

```
printf("Enter initial approximation=");
```

```
scanf("%f",&p);
```

```
printf("Enter number of iterations=");
```

```
scanf("%d",&n);
```

```
printf("Enter telerance=");
```

```
scanf("%f", &temp);
```

```
printf("Enter lower bound=");
```

```
scanf("%d", &delta);
```

```
for(i=1; i<n; i++)
```

```
{
```

```
printf("slope too small");
```

```
exit(1); }
```

```
r=p - m(p) / dm(p);
```

```
reller=fabs((r-p)/r);
```

```
p=r,
```

```
if((reller<=temp || fabs(m(r))== 0)).
```

```
{
```

```
printf("\n Approximation root=%f", r);
```

```
exit(1);
```

```
}
```

```
}
```

[A.10]

5. Find the root of equation $xe^x - 1 = 0$ between 0 and 1 using bisection method. Perform only two iterations. (2021)

$$f(0) = 0.e^0 - 1 = -1 < 0$$

$$f(1) = 2.e^1 - 1 = 1.7183 > 0$$

Hence a root of $f(x) = 0$ lies in between 0 and 1.

$$\therefore x_1 = (0+1)/2 = 0.5$$

$$f(0.5) = 0.5e^{0.5} - 1 = -0.1756$$

Hence the root lies in between 0.5 and 1

$$\therefore x_2 = (0.5+1)/2 = 0.75$$

Proceeding like this, we get the sequence of approximations as follows.

$$x_3 = 0.625$$

$$x_4 = 0.5625$$

$$x_5 = 0.59375$$

$$x_6 = 0.5781$$

$$x_7 = 0.5703$$

$$x_8 = 0.5664$$

$$x_9 = 0.5684$$

$$x_{10} = 0.5674$$

$$x_{11} = 0.5669$$

$$x_{12} = 0.5672$$

$$x_{13} = 0.5671$$

Hence, the required root correct to three decimal places is, $x = 0.567$.

6. Evaluate $\sqrt{12}$ to five decimal places by Newton's iterative method. (2021)

Newton's method says that

$$f(x) \approx l(x) = f(x_0) + f'(x_0)(x - x_0)$$

$\sqrt{12}$ can be found by the function $f(x) = x^2 - 12$

We know that $3 < \sqrt{12} < 4$, so $x_0 = 3$

$$f'(x) = 2x$$

1st Iteration :

$$f(x) = (3^2 - 12) + 2(3)(x - 3)$$

$$= -3 + 6(x - 3)$$

$$= -3 + 6x - 18$$

$$= 6x - 21$$

$$= 6x - 21 = 0$$

$$x = 21/6 = 7/2 = 3.5$$

Now we iterate with x_0 now being equal to $7/2$

2nd Iteration :

$$f(x) = ((7/2)^2 - 12) + 2(7/2)(x - 7/2)$$

$$= -1/4 + 7(x - 7/2)$$

$$= -1/4 + 7x - 49/2$$

$$= 7x - 99/4$$

$$7x - 99/4 = 0$$

$$x = 99/28 \sim 3.54$$

3rd Iteration :

$$f(x) = ((99/28)^2 - 12) + 2(99/28)(x - 99/28)$$

$$= 393/784 + 99/14(x - 99/28)$$

$$= 393/784 + 99/144(x - 9801/392)$$

$$= 99/14(x - 19209/784)$$

$$99/14 - 19209/784 = 0$$

$$x = 6403/1843 \approx 3.4648$$

4th Iteration :

$$f(x) = ((6403/1848)^2 - 12) + 2(6403/1848)(x - 6403/1848)$$

$$= 5.025 \cdot 10^{-3} + 6403/924(x - 6403/1848)$$

$$= 5.025 \cdot 10^{-3} + 6403x/924 - 24.01$$

$$= 6403 \times 924 - 24.005$$

$$6403 \times 924 - 24.005 = 0$$

$$x \approx 3.4641 \approx \sqrt{12}$$

7. Find the cubic polynomial in x , which takes on the values :

- 3, 3, 11, 27, 57 and 107

when $x = 0, 1, 2, 3, 4$ and 5 respectively.

(2019)

X	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
0	-3	6	2	6	0
1	3	8	8	6	0
2	11	16	14	6	0
3	27	30	20		
4	57	50			
5	107				

Since the fourth and higher order differences are zero. We have the required Newton's Interpolation formula

$$f(x_0 + ph) = f(x_0) + p\Delta f(x_0) + \frac{p(p-1)}{2} \Delta^2 f(x_0) + \frac{p(p-1)(p-2)}{6} \Delta^3 f(x_0) + \dots$$

Here $p = \frac{x - x_0}{h} = \frac{x - 0}{1} = x$

$$\Delta f(x_0) = 6, \Delta^2 f(x_0) = 2, \Delta^3 f(x_0) = 6$$

Putting all values in equation (1),

$$f(x) = -3 + 6x + \frac{x(x-1)}{2} \times 2 + x \frac{(x-1)(x-2)}{6} \times 6$$

$$f(x) = x^3 - 2x^2 + 7x - 3$$

is the required cubic polynomial.

(Ans)

8. Find a real root of:

$$f(x) = x^2 - x + 1 = 0$$

by Bisection method.

(2019)

Given;

$$f(x) = x^2 - x + 1 = 0$$

$$x^2 - x + 1 = 0$$

Using Shreedharcharya formula

$$d = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{(+1) \pm \sqrt{1-4}}{2}$$

$$= \frac{1 \pm \sqrt{-3}}{2} \quad (\text{imaginary root})$$

Hence can't be solved using bisection method.

(Ans.)

9. Find a root of:

$$x \log_{10} x - 1.2 = 0$$

By Newton - Raphson's method correct to three decimal places. (2019)

$$\text{Let } f(x) = x \log_{10} x - 1.2 \quad \dots(1)$$

$$f(1) = -1.2 = (-ve)$$

$$f(2) = 2 \log_{10} 2 - 1.2 = 0.59794 = (-ve)$$

$$f(3) = 3 \log_{10} 3 - 1.2 = 1.4314 - 1.2 = 0.23136 = (+ve)$$

So, $-f(x) = 0$ lies between 2 and 3, let $x_0 = 2$

$$f'(x) = \log_{10} x + x \frac{1}{x} \log_{10} e = \log_{10} x + 0.43429$$

$\therefore$ Now apply Newton's Raphson :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = \frac{0.43429x_n + 1.2}{\log_{10} x_n + 0.43429} \quad \dots(2)$$

Put $n = 0$, the first approx,

$$x_1 = \frac{0.42429 \times x_0 + 1.2}{\log_{10} x_0 + 0.43429} = \frac{0.86858 \times 1.2}{0.30103 + 0.43429}$$

$$= \frac{1.042296}{0.73532} = 2.81$$

Put $n = 1, 2$ in (2), we get

$$x_2 = \frac{0.43429 \times 2.81 + 1.2}{\log_{10} 2.81 + 0.43429} = 2.741$$

$$x_3 = \frac{0.43429 \times 2.741 + 1.2}{\log_{10} 2.741 + 0.43429} = 2.74064$$

Hence, $x_2 = x_3$ hence the required roots 2.74064

(Ans.)

10. Find the function whose first difference is e^x . (2018)

We know that, $\Delta e^x = e^{x+h} - e^x(e^h - 1)$. Where h is the interval of differencing

$$\text{Or } e^x = \frac{1}{e^h - 1} \Delta e^x$$

$$\text{Or } \Delta \left[\frac{e^x}{e^h - 1} \right] = e^x$$

$$\text{Or } f(x) = \frac{e^x}{e^h - 1} \quad (\text{Ans.})$$

11. Define Descartes' rule for sign. (2016, 2017, 2018)

Descart's Rule of Signs

A method of determining the maximum number of positive and negative real roots of a polynomial. For positive roots, start with the sign of the coefficient of the lowest (or highest) power.

Count the number of sign changes as proceed from the lowest to the highest power, then n is the maximum number of positive roots. Furthermore, the number of allowable roots is $n, n - 2, n - 4$.

For example, $f(x) = x^7 + x^6 - x^4 - x^3 - x^2 + x - 1$ since, there are three sign changes. These are a maximum of three times possible positive roots. For negative roots starting with a polynomial $f(x)$, write a new polynomial $f(-x)$ with the signs of all odd powers reversed, while leaving the signs of the even powers unchanged.

Then proceed as before to count the number of sign changes n . Then n is the maximum number of negative roots.

For example $f(x) = x^7 + x^6 - x^4 - x^3 - x^2 + x - 1$ and compute the new polynomial $f(-x) = -x^7 + x^6 - x^4 + x^3 - x^2 - x - 1$. There are four sign changes, so there are a maximum of four negative roots.

12. Find the value of x when $x^3 - 6x + 4 = 0$ by Newton-Raphson method. (2018)

Let $f(x) = x^3 - 6x + 4$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^3 - 6x_n + 4}{3x_n^2 - 6} = \frac{(3x_n^3 - 6x_n) - (x_n^3 - 6x_n + 4)}{3x_n^2 - 6}$$

$$= \frac{3x_n^3 - 6x_n - x_n^3 + 6x_n - 4}{3x_n^2 - 6} = \frac{2x_n^3 - 4}{3x_n^2 - 6}$$

$f(0) = 4 = +ve$; $f(1) = -1 = -ve$

So a root lies between 0 and 1

$f(1)$ is nearer to 0. Therefore we take initial root as

$x_0 = 1$

$$x_1 = \frac{2x_n^3 - 4}{3x_n^2 - 6} = \frac{2 \times 1^3 - 4}{3 \times 1^2 - 6} = \frac{2 - 4}{3 - 6} = \frac{-2}{-3} = 0.66666$$

$$x_2 = \frac{2(0.66666)^3 - 4}{3(0.66666)^2 - 6} = 0.73016$$

$$x_3 = \frac{2(0.73015873)^3 - 4}{3(0.73015873)^2 - 6} = \frac{3.22145837 - 4}{4.40060469} = 0.73205$$

$$x_4 = \frac{2(0.73204903)^3 - 4}{3(0.73204903)^2 - 6} = \frac{3.21539602 - 4}{4.439231265} = 0.73205$$

The root is 0.73205 correct to 5 decimal places. (Ans.)

13. Find the value of x when $x^2 - 5x + 2 = 0$ by Newton Raphson method. (2017)

Here $f(x) = x^2 - 5x + 2$

$f'(x) = 2x - 5$

$f(4) = (4)^2 - 5 \times 4 + 2 = -2$

$f(5) = (5)^2 - 5 \times 5 + 2 = 2$

Root lies between $x = 4$ and $x = 5$ by Newton

Raphson method

$$\begin{aligned}
 x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\
 &= x_n - \frac{x_n^2 - 5x_n + 2}{2x_n - 5}
 \end{aligned}$$

Taking $x_0 = 4$ we have

$$\begin{aligned}
 x_1 &= 4 - \frac{4^2 - 5 \times 4 + 2}{2 \times 4 - 5} \\
 &= 4 + \frac{2}{3} \\
 &= \frac{14}{3} \\
 &= 4.66666
 \end{aligned}$$

$$\begin{aligned}
 x_2 &= \frac{14}{3} - \frac{\left(\frac{14}{3}\right)^2 - 70/3 + 2}{\frac{28}{3} - 5} \\
 &= \frac{14}{3} - \frac{196 - 210 + 18}{13 \times 3} \\
 &= \frac{178}{39} = 4.5641
 \end{aligned}$$

$$\begin{aligned}
 x_3 &= \frac{178}{39} - \frac{\left(\frac{178}{39}\right)^2 - 5 \times \frac{178}{39} + 2}{2 \times \frac{178}{39} - 5} \\
 &= \frac{178}{39} - \frac{31684 - 34710 + 3042}{39 \times 161}
 \end{aligned}$$

$$x_3 = \frac{28658}{6279}$$

$$x_3 = 4.5616$$

(An

14. Find a real root of the equation $x^3 - 2x - 5 = 0$ by the method of false position correct to three decimal places.

(2017)

$$f(2) = 1 \text{ and } f(3) = 16$$

Hence the root lies in between 2 and 3

$$\begin{aligned}
 x_1 &= \frac{af(b) - bf(a)}{f(b) - f(a)} \\
 &= \frac{2 \times 16 - 3 \times (-1)}{16 - (-1)} \\
 &= \frac{35}{17} \\
 &= 2.058823529
 \end{aligned}$$

$$f(x_1) = f(2.058823529) = -0.390799917 < 0.$$

Therefore the root lies between 2.058823529 and 3.

The second approximation as

$$\begin{aligned}
 x_2 &= \frac{2.058823529 \times 16 - 3 \times (0.390799917)}{16 - (-0.390799917)} \\
 x_2 &= 2.08126366
 \end{aligned}$$

Same as the next approximation as

$$x_3 = 2.089639211$$

So the required root is 2.089639211.

(Ans.)

15. ♦ Find the real root of the equation $x^3 - 4x - 9 = 0$, correct to four significant digits using Bisection method. (2016)

♦ Given that one root of the equation :

$$x^3 - 4x - 9 = 0$$

Lies between 2.625 and 2.75. Find the root correct to four significant digits, Using Bisection method.

Here, $x_1 = 2.625$ and $x_2 = 2.75$

$$f(x_1) = f(2.625) = (2.625)^3 - 4 \times 2.625 - 9 = -1.4121$$

$$f(x_2) = f(2.75) = (2.75)^3 - 4 \times 2.6875 - 9 = 0.7969$$

$$x_3 = (2.625 + 2.75) / 2 = 2.6875$$

$$f(x_3) = f(2.6875) = (2.6875)^3 - 4 \times 2.6875 - 9 = -0.3391$$

The first approximation to root is 2.6875.

$$\text{Since, } f(x_1) \times f(x_3) = (-1.4121) \times (-0.3391) > 0$$

Replace x_1 by x_3

Now, we take $x_1 = 2.6875$ and $x_2 = 2.75$

$$f(x_1) = f(2.6875) = (2.6875)^3 - 4 \times 2.6875 - 9 = -0.339$$

$$f(x_2) = f(2.75) = (2.75)^3 - 4 \times 2.75 - 9 = 0.7969$$

$$x_3 = (2.6875 + 2.75) / 2 = 2.7186$$

$$f(x_3) = f(2.7186) = (2.7186)^3 - 4 \times 2.7186 - 9 = 0.2182$$

The second approximation to root is 2.7186

$$\text{Since, } f(x_1) \times f(x_2) = -0.3391 \times 0.2181 < 0 \quad \text{repla}$$

x_2 by x_3 .

Subsequent iterations will produce the following approximations to root :

2.7031, 2.7109, 2.707, 2.7051, 2.7061, 2.7066 observed in the last iterations, they produce no change at the first few significant positions. We take $x = 2.706$

16. Find the root of the equation $f(x) = x^2 - 3x + 2$ using Newton-Raphson method. (2016)

Let $x_1 = 0$ first approximation

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$f(x) = x^2 - 3x + 2$$

$$f'(x) = 2x - 3$$

$$x_2 = 0 - \frac{2}{(-3)} = \frac{2}{3} = 0.6667$$

Similarly,

$$x_3 = 0.6667 - \frac{0.4444}{(-1.6667)} = 0.9333$$

$$x_4 = 0.9333 - \frac{0.071}{(-1.334)} = 0.9959$$

$$x_5 = 0.9959 - \frac{0.0041}{(-1.0082)} = 0.9999$$

$$x_6 = 0.9999 - \frac{0.0001}{(-1.0002)} = 1.0000$$

Since, $f(1.0) = 0$, the root closer to the point $x = 0$ is 1.000

17. Find the real square root of 30 by Regula-Falsi method. (2015)

$\sqrt{30}$ lies between 5 to 6.

The first approximation root

$$x = \sqrt{30}$$

$$x^2 = 30$$

$$x^2 - 30 = 0$$

$$f(x) = x^2 - 30$$

$$f(5) = (5)^2 - 30 = -5$$

$$f(6) = (6)^2 - 30 = 6$$

$$x^{(1)} = 5 + \frac{(6-5) \times 5}{5+6} = 5.4545$$

Further $f(5.4545) = (5.4545)^2 - 30 = -0.2484$. Since $f(5.4545)$ and $f(6)$, root lies $x = 5.4545$ and $x = 6$. The second approximation

$$x^{(2)} = 5.4545 + \frac{(6 - 5.4545) \times 0.2484}{0.2484 + 6} = 5.4762$$

Now $f(5.4762) = (5.4762)^2 - 30 = -0.01123$ lies between $x = 5.4762$ and $x = 6$. Third approximation

$$x^{(3)} = 5.4762 + \frac{(6 - 5.4762) \times 0.01123}{0.01123 + 6} = 5.4772$$

Thus the required value of the root is 5.477. (Ans.)

18. Find the real root of the equation $x^4 - x - 9 = 0$, correct to three places of decimal. (2015)

Here $f(x) = x^4 - x - 9$

$$f'(x) = 4x^3 - 1$$

By Newton Raphson method, we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = \frac{3x_n^4 + 9}{4x_n^3 - 1}$$

Take $x_0 = 2$, first approximation

$$x_1 = \frac{3x_0^4 + 9}{4x_0^3 - 1} = \frac{3 \times 2^4 + 9}{4 \times 2^3 - 1} = \frac{57}{31} = 1.839$$

Second approximation

$$x_2 = \frac{3 \times (1.839)^4 + 9}{4 \times (1.839)^3 - 1} = 1.826$$

Required root is ≈ 1.826

19. Find the real root of the equation $x^6 - x^4 - x^3 - 1$ by Regula-Falsi method, correct to three places decimal. (20)

$$f(x) = x^6 - x^4 - x^3 - 1$$

$$f(1) = -2, f(2) = 39$$

Root lies between $x = 1$ and $x = 2$.

$$x^{(1)} = 1 + \frac{(2-1) \times 2}{2 + 39} = 1.0488$$

$$f(1.0488) = (1.0488)^6 - (1.0488)^4 - (1.0488)^3 - 1 = -2.0327$$

Now root lies between $x = 1.0488$ and $x = 2$.

$$x^{(2)} = 1.0488 + \frac{(2-1.0488) \times 2.0327}{2.0327 + 39} = 1.0959$$

$$f(1.0959) = (1.0959)^6 - (1.0959)^4 - (1.0959)^3 - 1 = -2.0263$$

Now root lies between $x = 1.0959$ and $x = 2$.

$$x^{(3)} = 1.0959 + \frac{(2-1.0959) \times 2.0263}{2.0263 + 39} = 1.1406$$

$$\text{Now, } f(1.1406) = (1.1406)^6 - (1.1406)^4 - (1.1406)^3 - 1 = -1.9745$$

Now root lies between $x = 1.1406$ and $x = 2$.

$$x^{(4)} = 1.1406 + \frac{(2-1.1406) \times 1.9745}{1.9745 + 39} = 1.1820$$

Now,

$$f(1.1820) = (1.1820)^6 - (1.1820)^4 - (1.1820)^3 - 1 = -1.8762$$

Root lies between $x = 1.182$ and $x = 2$

$$x^{(5)} = 1.182 + \frac{(2-1.182) \times 1.8762}{1.8762 + 39} = 1.2195$$

$$f(1.2195) = (1.2195)^6 - (1.2195)^4 - (1.2195)^3 - 1 = -1.7361$$

$$\text{Now, } x^{(6)} = 1.2195 + \frac{(2 - 1.2195) \times 1.7361}{1.7361 + 39} = 1.2528$$

So the required root is 1.253.

(Ans.)

20. What are the properties of the equations? (2015)

Properties of Equations

Given that A , B and C represent real or complex algebraic expressions, then the following are true :

Addition Property of Equality : If $A = B$, then $A + C = B + C$. Note that since subtracting from both sides is the same as adding a negative amount to both sides, subtraction from both sides also is covered by this property.

Multiplication Property of Equality : $A = B$, then $AC = BC$.

Division Property of Equality : If $A = B$, then $A/C = B/C$ where $C \neq 0$.

Absolute Value Equation Property : If $|A| = B$, then $A = B$ and $-A = B$ are both possible solutions $\neq 0$.

Simple Quadratic Equations : If $A^2 = B$, and A is unknown, then $A = \pm\sqrt{B}$. Using this property is known as Extracting Square Roots.

General Quadratic Equations : If $ax^2 + bx + c = 0$ where a , b , and c are real coefficients and x is a variable, then

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This is known simply as the Quadratic Formula.

Radical Equations : If $\sqrt{A} = B$, and A is unknown, then $A = B^2$ is an equation that when solved results in possible solutions of A . This is known as eliminating the Radical by Squaring. We could also call this property

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Rational Exponent Equations : If $A^{M/N} = B$, and A is unknown and M is even, then $A = \pm B^{N/M}$ is an equation that when solved results in possible solutions of A . This is known as eliminating a Rational Exponent.

This method also applies to solving n th root radical equations. For example, to solve an equation with a fourth root radical, take the 4th power of both sides.

Rational Exponent Equations : If $A^{MN} = B$, and A is unknown and M is odd, then $A = B^{\frac{1}{M}}$ is an equation that when solved results in possible solutions of A . This is known as eliminating a Rational Exponent. This method also applies to solving n th root radical equations. For example, to solve an equation with a third root radical, take the 3rd power of both sides.

Zero Product Law : If $XY = 0$, then $X = 0$ or $Y = 0$ or both X and $Y = 0$. Note : X and Y are often factors. So for example If $(x + 3)(2x - 1) = 0$, then we can conclude that $x + 3 = 0$ or $2x - 1 = 0$ and we can solve each equation to find all solutions for x .

The number of positive roots of a polynomial with real coefficients is equal to the number of "changes of sign" in the list of coefficients, or is less than this number by a multiple of 2.

Since the negative roots of the polynomial equation $f(x) = 0$ are positive roots of the equation $f(-x) = 0$, the rule can be readily applied to help count the negative roots as well.

21. Using Newton-Raphson method evaluate to two decimal figures, the root of the equation $e^x = 3x$ lying between 0 and 1. (Dec. 2013)

$$f(x) = e^x - 3x$$

$$f'(x) = e^x - 3$$

$$\text{Also } f(0) = e^0 - 3 \times 0 = 1 - 0 = 1$$

$$f(1) = e^1 - 3 \times 1 = 2.718 - 3 = -0.282$$

Hence root of given equation lies between 0 to 1.
Newton Raphson method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Now taking $x_0 = 0$ we have

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 0 - \frac{1}{-2} = \frac{1}{2} = 0.5$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 0.5 - \frac{0.1487}{(-1.3513)} = 0.61$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 0.61 - \frac{0.0104}{(-1.1596)} = 0.619$$

So the root evaluated to two decimal places is 0.61.

22. Find the real root of $3x - \cos x - 1 = 0$ by the method of False position. (Dec. 2013)

$$f(x) = 3x - \cos x - 1$$

$$\text{Now } f(0) = 3 \times 0 - \cos 0 - 1 = -2$$

$$f(1) = 3 \times 1 - \cos 1 - 1 = 1.4597$$

So the root lies between 0 and 1. Now

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)} = \frac{0 \times 1.4597 - 1 \times (-2)}{1.4597 + 2}$$

$$= \frac{2}{3.4597} = 0.5781$$

$$f(x_3) = f(0.5781) = 3 \times 0.5781 - \cos 0.5781 - 1 = -0.1032$$

Thus, the first approximation root is 0.5781

$$\text{Since } f(x_1) \times f(x_3) = -2 \times -0.1032 > 0$$

The new search interval (0.5781, 1)

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)} = \frac{0.5781 \times 1.4597 - 1 \times (-0.1032)}{1.4597 + 0.1032}$$

$$= \frac{0.9471}{1.5629} = 0.6060$$

$$f(x_3) = 3 \times 0.6060 - \cos 0.6060 - 1 = -0.0041$$

$$\text{Since } f(x_1) \times f(x_3) = (-0.1032) \times (-0.0041) > 0$$

The new search interval (0.6060, 1)

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)} = \frac{0.6060 \times 1.4597 - 1 \times (-0.0041)}{1.4597 + 0.0041}$$

$$= 0.6071$$

Thus, the third approximation to the root is 0.6071.s

$$f(x_3) = f(0.6071) = 3 \times 0.6071 - \cos 0.6071 - 1 = -0.0002$$

Since; $f(x_1) \times f(x_3) = -0.0041 \times (-0.0002) > 0$

The new search interval (0.6071, 1)

$$x_3 = \frac{x_1 f(x_2) - x_2 f(x_1)}{f(x_2) - f(x_1)} = \frac{0.6071 \times 1.4597 - 1 \times (-0.0002)}{1.4597 + 0.0002}$$
$$= 0.6072$$

$$f(x_3) = 3 \times 0.6072 - \cos 0.6072 - 1 = -0.0001$$

Thus, the fourth approximation to the root is 0.6072. Since last two iterations produce no change at the first three decimal positions. Therefore, we take $x = 0.607$ as the desired solution correct to three decimal positions.

□□